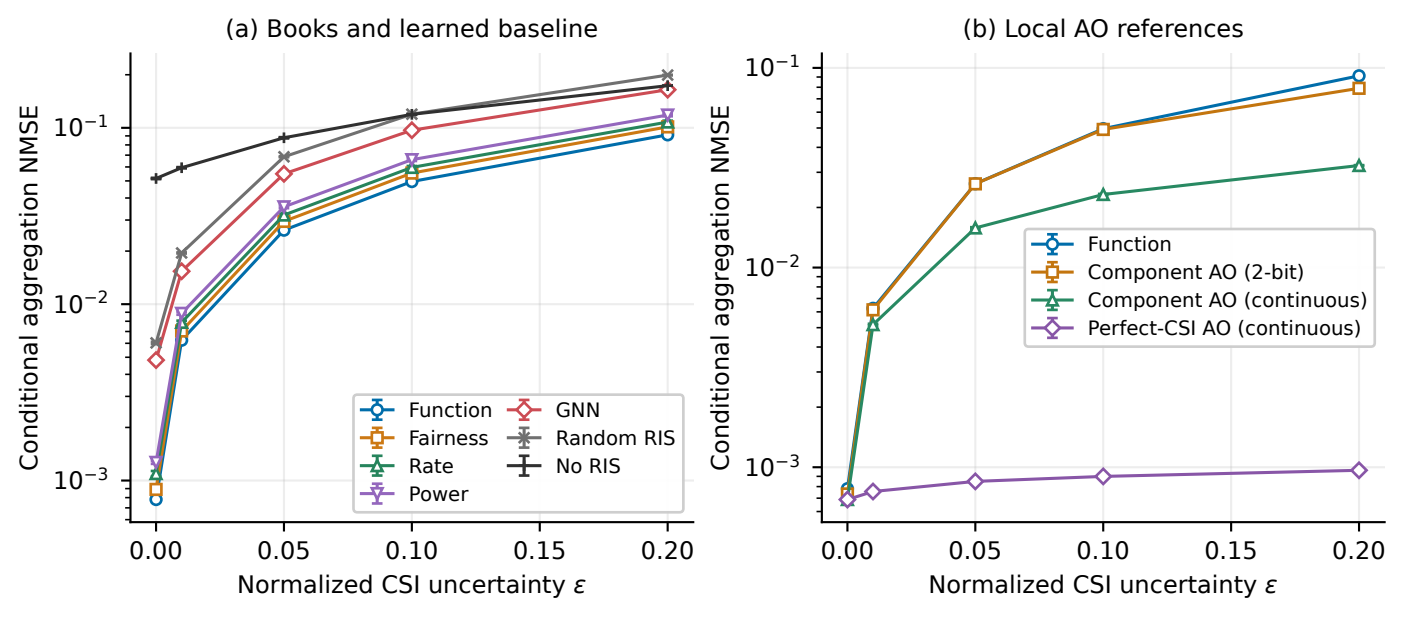
(a) Books and learned baseline
Conditional aggregation NMSE
10^{-1}
10^{-2}
10^{-3}
0.00
0.05
0.10
0.15
0.20
Normalized CSI uncertainty ε
Function
Fairness
Rate
Power
GNN
Random RIS
No RIS
(b) Local AO references
Conditional aggregation NMSE
10^{-1}
10^{-2}
10^{-3}
0.00
0.05
0.10
0.15
0.20
Normalized CSI uncertainty ε
Function
Component AO (2-bit)
Component AO (continuous)
Perfect-CSI AO (continuous)